

All My Work

Your saved learning record for Molecular Genetics.

1 completed activity run. Required checks and formative practice are recorded separately.

Unfinished work: 0 required check(s), 1 practice run(s). Resume them from their original sections; they are not counted as completed.

Writing and flash-card self-assessments are not automatically graded. Completion is not a mastery certificate.

Mixed practice

9/18/2026, 12:21:08 AM • Elapsed 0m 1s • Formative practice

Review submitted responses and attempts

First-try result: 7/7. Later corrections are recorded separately.

What does letter D identify?

Options shown: Sugar–phosphate backbone • 3' end • Complementary base pair

Attempt 1: 3' end — Correct.

Reference: 3' end

Identify the term illustrated: Different labels followed phage DNA and protein in separate preparations.

Attempt 1: radioactive tracer — Correct.

Reference: radioactive tracer

A new bacterial lineage contains a resistance allele identical to one in another population. Does this alone prove a new mutation occurred in the recipient?

Options shown: No; because bacteria have no genetic material · Yes; every inherited trait requires a new mutation · No; transfer of an existing allele is another explanation · Yes; DNA cannot move between bacteria

Attempt 1: No; transfer of an existing allele is another explanation — Correct.

Reference: No; transfer of an existing allele is another explanation

Correct this incorrect statement: Grouping begins at any nucleotide without affecting the result.

Self-report: Study again.

Reference: The grouping of a nucleotide sequence into successive triplets for translation. The position of the start establishes which triplets are read.

Select the TWO accurate statements about solving dna, rna and protein problems.

Options shown: Complementing letters without tracking direction can reverse the message. · An amino-acid sequence is a series of RNA bases. · A practice fragment must state whether it is a complete message or an internal segment. · If a sequence has the right length, its bases and direction must be correct.

Attempt 1: true-1 → true-2 — Correct.

Reference: Complementing letters without tracking direction can reverse the message. · A practice fragment must state whether it is a complete message or an internal segment.

Put these steps in order: Transcription and RNA processing

Options shown: Introns are removed and exons joined · Processed mRNA leaves the nucleus · Complementary pre-mRNA is synthesized · A ribosome can translate the message · RNA polymerase begins transcription

Attempt 1: RNA polymerase begins transcription → Complementary pre-mRNA is synthesized → Introns are removed and exons joined → Processed mRNA leaves the nucleus → A ribosome can translate the message — Correct.

Reference: RNA polymerase begins transcription → Complementary pre-mRNA is synthesized → Introns are removed and exons joined → Processed mRNA leaves the nucleus → A ribosome can translate the message

Which term matches this description? A shift in reading frame caused by an insertion or deletion not divisible by three in a coding region.

Options shown: frameshift · nonsense mutation · silent substitution · mutation

Attempt 1: frameshift — Correct.

Reference: frameshift

Which term fits this definition? RNA that forms structural and catalytic parts of ribosomes.

Self-report: Study again.

Reference: ribosomal RNA

Name the term described: An enzyme that extends a new DNA strand by adding nucleotides to its 3' end.

Attempt 1: DNA polymerase — Correct.

Reference: DNA polymerase

Explain conjugation.

Self-report: Study again.

Reference: Transfer of genetic material between bacterial cells through direct contact.

Fraye work

transformation — Draft

Meaning: An edited explanation.

Key features: A specific learner explanation 2.

Example: A specific learner explanation 3.

Non-example / common confusion: A specific learner explanation 4.

hereditary material — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

radioactive tracer — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

controlled comparison — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

nucleotide — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

complementary base pairing — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

antiparallel — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

Chargaff's relationships — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

Collected optional responses

Biotechnology and unintended consequences

The comparison alone cannot isolate the proposed chemical effect. Hormone production and receptor response are distinct steps.

Textbook question work

Draft and saved question responses. Optional and ungraded.

No textbook question work yet.

Supported and independent practice